

Autonomous Environment Exploration using TurtleBot3 in Gazebo

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I. PROBLEM STATEMENT

The task is to demonstrate an algorithm capable of exploring an environment with obstacles present. The final goal is to build a map of the environment. Teleoperation is not allowed and all navigation and planning must happen autonomously.

In unknown environments, a mobile robot must rely entirely on its onboard sensors to perceive obstacles, identify unexplored regions, and navigate safely through the environment. The challenge is to enable the robot to autonomously explore the environment while progressively constructing a map of the explored space using the TurtleBot3 platform in the Gazebo simulation environment.

II. INTRODUCTION

Autonomous exploration is an important capability in mobile robotics, especially when robots operate in environments where no prior map of the surroundings is available. In such cases, the robot must rely on its onboard sensors to observe the environment, detect obstacles, and gradually build a representation of the space while navigating through it. This allows the robot to move safely while simultaneously learning about the environment.

Such capabilities are useful in several practical situations. For example, robots used in search and rescue operations may need to explore unfamiliar environments, and mobile robots in warehouses often navigate large spaces where efficient mapping and navigation are required.

Robotic frameworks like ROS2 make it easier to develop systems that combine sensing, mapping, and navigation. Simulation tools such as Gazebo also provide a convenient way to test robotic algorithms in a controlled environment before applying them to real robots.

In this project, an autonomous exploration system is developed using the TurtleBot3 robot in the Gazebo simulation environment. The robot uses LiDAR sensor data to observe the environment and applies simultaneous localization and mapping (SLAM) to build a map while navigating through the space. An exploration algorithm is used to identify unexplored regions of the environment and guide the robot toward those areas.

III. TOOLS USED

The development of the autonomous exploration system relies on several robotics software tools and frameworks that provide simulation, mapping, navigation, and visualization capabilities. These tools enable the robot to perceive its environment, build maps, plan paths, and move autonomously.

ROS2 Jazzy

ROS2 is the main software framework used to connect all parts of the robot system. It allows different programs such as mapping, navigation, and sensor processing to share data and work together.

Gazebo Simulator

Gazebo is the simulation environment where the TurtleBot3 robot is tested. It creates a virtual world with walls and obstacles and also simulates sensors like LiDAR and odometry.

TurtleBot3 Burger

TurtleBot3 Burger is the robot model used in this project. It has wheels for movement and a LiDAR sensor that helps the robot detect obstacles and understand its surroundings.

SLAM Toolbox

SLAM Toolbox is used to build a map of the environment while the robot is moving. It uses LiDAR data to find the robot's position and slowly create a map of the area.

Navigation2 (Nav2)

Navigation2 is responsible for helping the robot move safely in the environment. It calculates a path to the goal and controls the robot so that it avoids obstacles.

RViz2

RViz2 is a visualization tool used to see what the robot is doing. It shows the robot model, sensor data, and the map that is being created during exploration.

IV. SYSTEM WORKFLOW AND EXECUTION

This section explains how the complete exploration system works and how different components interact with each other. The system runs several ROS2 nodes at the same time in different terminals. Each node performs a specific task such as running the simulation, creating the map, controlling robot movement, or selecting the next exploration target.

In this system, the robot begins in a simulated environment and starts collecting sensor data. Using this data, it builds a map of the surroundings, identifies unexplored areas, and

moves toward those areas until the whole environment has been explored.

A. Step 1: Launching the Gazebo Simulation

The first step is to start the Gazebo simulation. In the first terminal, the TurtleBot3 Gazebo world is launched using a ROS2 command.

When Gazebo starts, the TurtleBot3 robot appears inside the virtual environment. The robot can move inside this world just like a real robot. The simulator also handles the robot movement and collisions with obstacles.

The robot in the simulation is equipped with a LiDAR sensor and wheel encoders that allow it to observe the environment and track its motion.

B. Step 2: Sensor Data Generation

After the simulation starts, the robot sensors begin producing data continuously. This data is published as ROS2 topics so that other parts of the system can use it.

Some of the main topics generated by the robot are:

- **/scan** – LiDAR data used to measure distances to nearby obstacles.
- **/odom** – wheel odometry that estimates how the robot has moved.

These sensor readings update continuously as the robot moves in the environment.

C. Step 3: Mapping Using SLAM Toolbox

In another terminal, the SLAM Toolbox node is launched. This node is responsible for creating a map of the environment while the robot is moving.

SLAM Toolbox receives the LiDAR scan data from the **/scan** topic and the robot motion information from the **/odom** topic. Using this information, it gradually builds a two-dimensional occupancy grid map of the environment.

The generated map is published on the **/map** topic and is updated continuously as the robot discovers new areas.

In this map, each cell represents a part of the environment and can be classified as:

- Free space
- Occupied space (obstacles)
- Unknown space

The map can also be viewed in real time using RViz2.

D. Step 4: Launching the Navigation Stack

In another terminal, the Navigation2 stack is started. This system is responsible for planning paths and controlling the robot's movement.

Navigation2 uses the map generated by SLAM to find safe paths in the environment. It also keeps track of the robot's current position using odometry information.

When a navigation goal is given, Nav2 calculates a path to that location and sends movement commands to the robot.

These commands are published to the **/cmd_vel** topic, which controls the robot's motion inside the Gazebo simulation.

E. Step 5: Frontier Detection and Exploration

Once the mapping and navigation systems are running, the custom frontier exploration node is launched.

This node continuously checks the map generated by SLAM Toolbox. It looks for frontier regions, which are the boundaries between explored space and unexplored areas.

These frontier locations represent places where the robot can move in order to discover new parts of the environment.

The exploration node selects one of these frontier points and sends it as a navigation goal to the Navigation2 system.

F. Step 6: Robot Navigation and Map Update

After a frontier goal is sent, Navigation2 calculates a path and moves the robot toward that location.

While the robot moves, the LiDAR sensor keeps scanning the surroundings. This new data is used by SLAM Toolbox to update the map and add the areas that the robot has just discovered.

Because the map keeps changing, new frontier regions appear during the exploration process.

G. Step 7: Continuous Exploration Process

The exploration system works in a repeating cycle:

- 1) Detect frontier regions in the map
- 2) Select one frontier as the next goal
- 3) Navigate the robot toward that frontier
- 4) Update the map using new sensor data
- 5) Detect new frontiers

This cycle continues automatically as the robot explores more parts of the environment.

H. Step 8: Completion of Exploration

Eventually, the system reaches a stage where no more frontier regions can be found. This means that all reachable areas of the environment have been explored.

At this point, the exploration process stops, and the final map represents the complete layout of the environment.

V. SYSTEM ARCHITECTURE

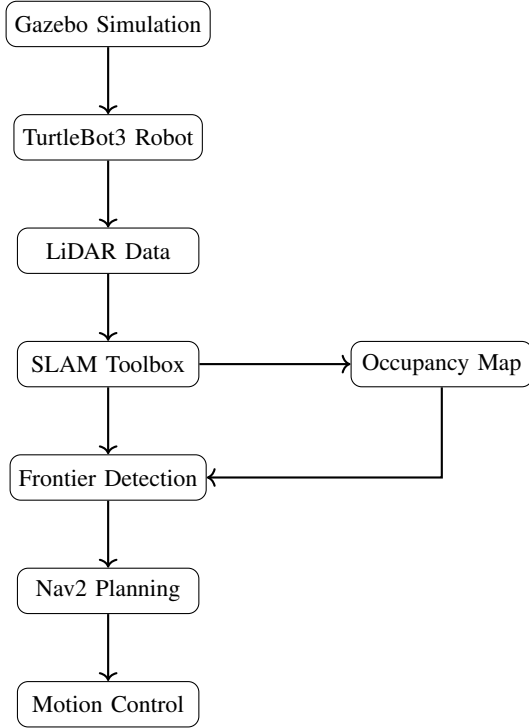


Fig. 1. System architecture of the autonomous exploration framework.

VI. EXPLORATION ALGORITHM

The exploration strategy implemented in this project is based on the frontier exploration approach. Frontier exploration is an efficient method for guiding a robot through unknown environments. The key idea behind this method is to identify the boundary between explored and unexplored regions in the occupancy grid map and use these boundaries as exploration targets.

Initially, the robot starts with no prior knowledge of the environment. As the robot begins scanning the surroundings using its LiDAR sensor, the SLAM Toolbox generates an occupancy grid map. This map consists of three types of cells: free space, occupied space, and unknown space. The frontier exploration algorithm analyzes this map to detect frontier cells.

A frontier is defined as a free cell that is adjacent to an unknown cell in the map. These frontier cells represent potential areas where the robot can move to discover new parts of the environment. Once the frontier cells are detected, the algorithm groups them into clusters and determines the most suitable frontier to explore next.

The selection of the frontier is typically based on criteria such as distance from the robot or accessibility. In this project, the robot selects the nearest reachable frontier as the exploration goal. The coordinates of the selected frontier are then sent to the Navigation2 stack as a navigation goal.

After receiving the goal, the Navigation2 stack computes a collision-free path from the robot's current position to the frontier location. The robot then follows this path while

continuously updating the map using SLAM. When the robot reaches the selected frontier, new areas of the environment become visible and additional frontiers are detected. This exploration process repeats until no more reachable frontiers remain in the environment.

VII. EXPERIMENTAL SETUP

The experiments for this project were conducted using the Gazebo robotics simulator integrated with ROS2. The TurtleBot3 Burger robot model was used as the mobile robot platform. The robot is equipped with a simulated LiDAR sensor that provides range measurements of the surrounding environment.

The exploration environment was created using Gazebo world files that include walls and obstacles representing an indoor environment. The robot was placed at an initial starting position and allowed to explore the environment autonomously.

The SLAM Toolbox package was used to generate an occupancy grid map of the environment using LiDAR data. The Navigation2 stack was configured to perform global and local path planning for safe navigation. The frontier exploration node continuously analyzed the generated map and provided navigation goals for exploration.

RViz was used to visualize the robot's position, laser scan data, and the occupancy grid map during exploration. This allowed real-time observation of how the robot navigated through the environment and how the map expanded over time.

VIII. RESULTS AND OBSERVATIONS

The implemented system successfully demonstrated autonomous exploration in the simulated environment. Once the exploration node was activated, the robot began scanning the environment using the LiDAR sensor and generating a map using the SLAM Toolbox.

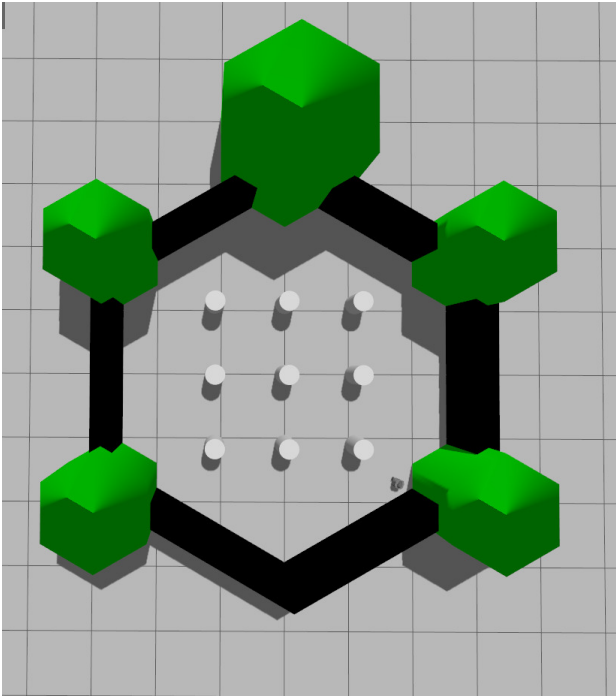


Fig. 2. Gazebo simulation environment showing the TurtleBot3 robot placed inside the obstacle environment.

As the map was gradually built, the frontier exploration algorithm identified unexplored regions and generated navigation goals accordingly. The robot autonomously moved toward these frontiers while avoiding obstacles using the Navigation2 stack. The map expanded progressively as the robot discovered new regions of the environment.

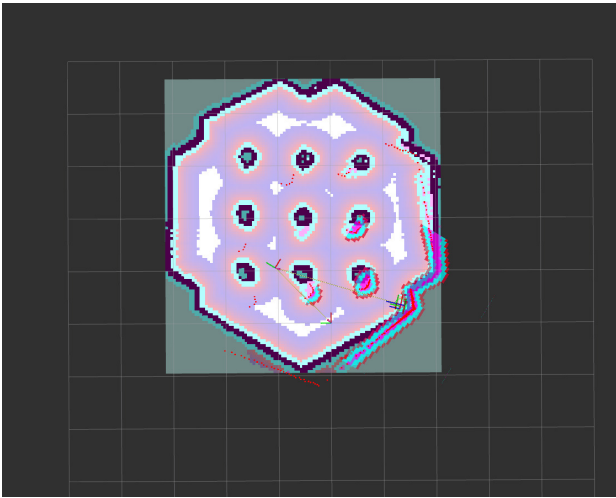


Fig. 3. RViz visualization showing the robot exploring the environment and updating the occupancy grid map.

The exploration process continued until all reachable frontiers were explored. At this stage, the environment was completely mapped and the exploration algorithm terminated automatically.

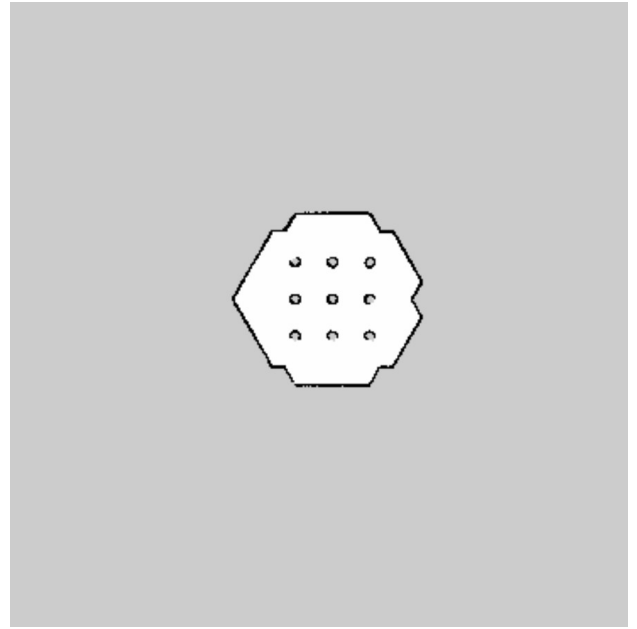


Fig. 4. Final occupancy grid map generated by SLAM after exploration.

The final result was an occupancy grid map representing the layout of the environment including walls and free space. The results confirm that the integration of SLAM Toolbox, Navigation2, and frontier exploration algorithms can successfully enable autonomous exploration in a simulated robotic environment.

IX. CONCLUSION

In this project, an autonomous exploration system was successfully implemented using the TurtleBot3 Burger robot in a simulated Gazebo environment. The primary objective was to enable the robot to explore an unknown environment containing obstacles and generate a map without any human intervention.

The system integrates several important robotics components including SLAM Toolbox for mapping, Navigation2 for path planning and obstacle avoidance, and a frontier-based exploration algorithm for selecting exploration targets. By combining these components within the ROS2 framework, the robot was able to autonomously navigate through the environment and incrementally build an occupancy grid map.

The results demonstrate that the implemented system can effectively explore an unknown environment, avoid obstacles, and generate an accurate representation of the surroundings. This project highlights the capabilities of modern robotics frameworks such as ROS2 and demonstrates how multiple robotic subsystems can be integrated to achieve autonomous behavior.